

AN EXPERIMENTAL DESIGN FOR EVALUATING SONAR DETECTION PERFORMANCE

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Key words: D-optimality; experimentation; logistic; minimax; regression; sonar search.

Summary: Recently we planned and carried out an experiment for evaluating the detection performance of a sonar system. This experiment was designed with little prior knowledge and was primarily based on practical statistical and other considerations and the expertise of sonar engineers. Subsequently we used the results obtained from the above-mentioned experiment to modify the experimental layout and improved its efficiency in a D-optimal sense. This paper is focused on the experimental design process and will guide the reader through our reasoning, from the initial design considerations to the final proposal. The result is a design which meets the practical requirements and has an efficiency which is close to the optimal, albeit impractical one.

1. INTRODUCTION

The search for underwater objects occurs frequently in both civilian and military contexts. Examples include the locating of sunken ships or aircraft that have crashed into the sea as well as detecting, for example, submarines and underwater mines.

A search strategy for a specific object in a given area may be subdivided into a number of phases. In the first phase, the area is scanned in a systematic manner, for example, searching in parallel tracks using sonar. Once an object has been detected, it is then classified as being either similar

to the object of interest or dissimilar to it. If it is similar then it is closely inspected, using divers or an underwater television camera or both, to ascertain its exact nature. However, a successful search strategy can only be devised if the capabilities of the equipment are known. How far can one see with the naked eye, how good is sonar at detecting objects under various conditions and what are the capabilities of the television cameras and divers? These are questions that will affect the search strategy.

In this paper we focus on the question of designing an experiment to evaluate the detection performance of the sonar equipment. Our broader aim is to develop effective search strategies and in particular to determine a sensible track spacing in a parallel track search using sonar.

A sonar system is rather complex and experimenting with it in the ocean – an unfriendly and extremely variable environment subject to many uncontrollable factors – called for very careful planning. Therefore a substantial amount of time was spent in gaining familiarity with the system and ascertaining how it is used in practice. This meant spending many hours with sonar engineers and with the client on board a ship. This work has again emphasized to us a point made in a number of papers discussing the role of the statistician in experimentation, see for example Hunter (1981), Bishop, Peterson and Trayser (1982), Hahn (1984a) and the review papers by Steinberg and Hunter (1984) and by Coleman and Montgomery (1993). In these papers the important role of the statistician in the planning and design process is mentioned and especially the fact that often his major contribution is made during the planning stages of the experiment prior to formal statistical design and data analyses. Furthermore, it is often felt, see Hahn (1977), (1984a), that proper experimental planning and design is more important than sophisticated data analysis. We also note that according to Hahn (1984b) two competent statisticians, faced with the same problem, will likely propose two quite different, yet quite reasonable, designs.

We first designed a pilot experiment and then used the results to improve the efficiency of the experimental design in a D-optimal sense. The layout of this paper is as follows. In the next section a short introduction to

sonar is given followed by a discussion of some relevant concepts in sonar search. In section 3 the design of the initial experiment is discussed as well as the typical design considerations. Assuming a logistic regression model for the detection probability curves and using the criteria of local D-optimality we show in section 4, how the efficiency of the initial design can be improved. This resulted in a final design having efficiency quite close to that of the optimal (but rather impractical) one. Section 5 contains some concluding remarks.

2. SONAR AND THE QUANTIFICATION OF SONAR SEARCH

In this section we first give a very brief discussion of sonar and the factors which influence sonar detection performance. Then we discuss the lateral range function and show how it can be used to evaluate sonar detection capability. We conclude with the important role played by the lateral range function in choosing the track spacing in a parallel track search plan.

2.1 Sonar detection

Sonar is an acronym for Sound Navigation and Ranging and designates that branch of applied acoustics which studies the propagation of acoustic energy through water. Systems which use underwater acoustic energy for observation or communications are referred to as sonar systems. Although electromagnetic waves, as used, for example, in radio and radar, are propagated with extreme rapidity and through great distances in certain media, sea water is essentially impervious to them at most frequencies. Acoustic pressure waves, on the other hand, are capable of being transmitted in the sea through significant distances. Thus sonar is the primary sensing system used for underwater work. We are primarily interested in the detection role of sonar. The fundamental problem in any detection system is to find some way of separating information indicating the presence of the object to be detected from the background noise caused by similar pieces of information completely unrelated to the object to be detected. Specifically, a sonar system attempts to do this using acoustic energy either radiated by or

reflected from a target. These are respectively referred to as passive and active sonar. We are only concerned with the latter type.

The transmission of sound through the sea is influenced by such factors as depth, salinity and temperature and it is thus important to measure these factors during experimentation. Also, when a sound beam strikes a target, the amount reflected depends on the size and shape of the target, the nature of the target material and its orientation. See Urick (1975) or Coppens et al. (1979) for further details.

The overall aim of the quantification of sonar search is to determine how search equipment should be used in a particular situation to give the best result. Clearly the evaluation of the sonar detection capability is an important contribution to this overall aim. We will quantify this evaluation in terms of the so-called lateral range function which we now discuss.

2.2 Lateral range and the lateral range function

Since the primary aim of the experiments was to evaluate the detection capability of the sonar system, one can think of it as part of a search problem: Search a specified area for all underwater objects of a specific type. Effectiveness would be partially measured by the fraction of objects detected. We will consider a so-called parallel track search plan in which an area is searched using equidistant parallel tracks. (See section 2.3.) This is a common method of search used in a situation where no prior knowledge is available about the location of the target(s) within the area to be searched. The shortest (perpendicular) distance between the search object and the sonar track is called the *lateral range*. The point at which the perpendicular dropped from the search object intersects the track is called the closest point of approach. Note that although detection can occur at any point along a track, the lateral range provides a convenient measure of distance between the search object and the sonar. This is illustrated in figure 1 below:

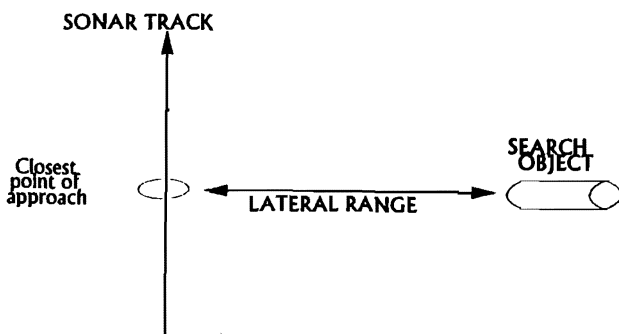


Figure 1: Illustration of lateral range

The *lateral range function* at x is now defined to be the probability that a search along the track will detect an object situated at lateral range x from it. We assume the track of the searcher is infinitely long in both directions, where by the latter is meant a line long enough such that the probability of detection would not increase if it was made longer (see for example Washburn, 1981).

Remarks:

- (i) Note that the specific point at which detection occurs is not relevant; the lateral range function simply gives the probability of detecting the object somewhere on the track.
- (ii) Lateral range functions usually decrease with increasing range. Furthermore, it is usually assumed that the lateral range function is zero for lateral ranges beyond some finite value d .
- (iii) Note that there are many factors determining the lateral range function, for example prevailing environmental conditions (temperatures, salinity, etc.), the acoustic properties of the target, its aspect, etc.

2.3 A parallel track search plan

An important question in planning a search operation is what track spacing to use for a specified "success probability". We now discuss the important role played by the lateral range function in answering this question.

Consider the search of a rectangular area with parallel tracks spaced at a distance s apart. We assume that the position of a target is uniformly distributed over the search area and that the latter is covered by $2N + 1$ tracks as shown in figure 2.

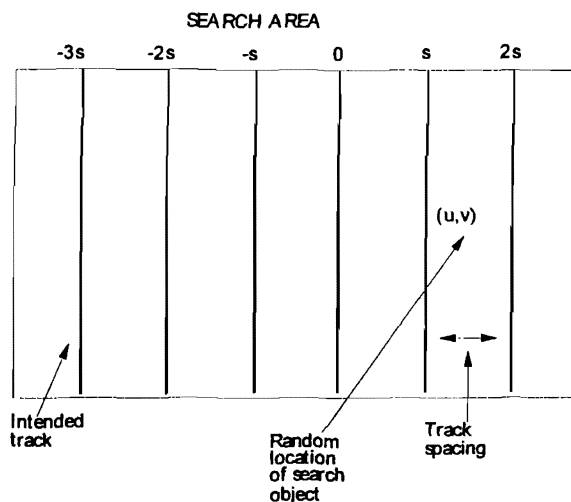


Figure 2: Intended track placement for parallel line search

Denote the lateral range function by $f(\cdot)$ and the position of the target by (u, v) . If the ship is on track js , then the range at the closest point of approach is $u - js$. If we assume that detections on different tracks occur independently, it follows that

$$\begin{aligned}
 (1) \quad p(u, s) &\equiv P(\text{at least one detection} \mid \text{target at } (u, v)) \\
 &= 1 - \prod_{j=-N}^N (1 - f(|u - js|)).
 \end{aligned}$$

Thus, for a uniform distribution of the target over the search area, we get

$$\begin{aligned}
 (2) \quad P_D(s) &\equiv P(\text{detecting the target using track spacing } s) \\
 &= (2Ns)^{-1} \int_{-Ns}^{Ns} p(u, s) du \\
 &= (2Ns)^{-1} \int_{-Ns}^{Ns} \left[1 - \prod_{j=-N}^N (1 - f(|u - js|)) \right] du.
 \end{aligned}$$

Now, for a particular search exercise we can calculate the detection probability once we have the lateral range function $f(\cdot)$. For a specified probability one can then use the above expression for $P_D(s)$ to decide which track spacing to use to "achieve" this specified search success. In order to do this, the lateral range function must be known. The experiments described in what follows, enable one to estimate a lateral range function assuming a logistic model.

3. FIRST EXPERIMENTAL PLAN AND DESIGN CONSIDERATIONS

To bring the problem within practical bounds we were interested in the detection capability of a particular sonar for targets similar to 200 litre *metal* drums as well as *non-metal* targets of a similar size and shape, placed on a sandy sea bed at a depth of about 20 metres. The designs proposed in this paper incorporated both types of objects. This made possible the estimation of both detection probability curves from the same experiment. The experiments consisted of placing a number of objects, called targets, in a suitable area and then running a number of tracks over this area, each time noting objects detected as well as their positions. We now describe this in more detail.

Based on previous sonar records, a suitable region which was relatively free from artifacts, was chosen in False Bay. Its area was one square nautical mile and was oriented north to facilitate navigation. It was divided into four blocks of equal size, four targets being placed in each block, giving a total of sixteen targets (see figure 3). These targets were chosen to have roughly the

same shape and volume as the types of target that we were primarily interested in. Of the sixteen targets half were made of metal while the remaining were non-metal objects.

Some prior information, obtained from sonar engineers, about the sonar system under investigation indicated that only rarely would detection be successful at lateral ranges beyond 200 metres. For this reason the observations had to be fairly uniformly spread over the 0 to 200 metre range. In addition, to verify the above prior information, some observations had to be made in the 200 to 600 metre range. In view of this and of the sparsity of 16 points in an area of one square nautical mile, it was decided to divide the area into 4 blocks of equal size with 4 targets in each block. The number of targets and the size of the area were chosen so that the density of targets would not appear to the sonar operators to be obviously higher than is normal in the False Bay area. It was decided to place objects relatively far from the origin (the centre of the search area) into diagonal blocks and relatively close to the origin in the other two diagonal blocks. Call the first two blocks (distant targets) Q1 and Q3 and the second two (close targets) Q2 and Q4. This is illustrated in figure 3 below.

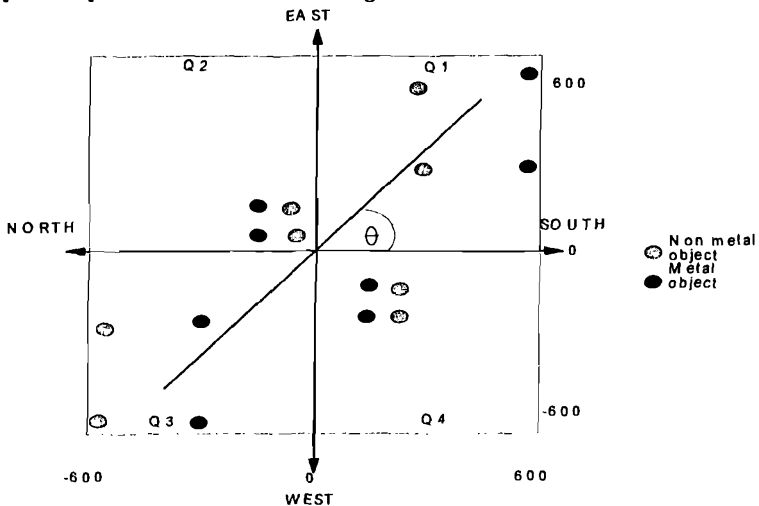


Figure 3: Experimental layout

Runs were to be made only between Q1 and Q3. This enabled better control of the distribution of observations over the various lateral ranges. Runs between Q2 and Q4 result in too large a concentration of observations at close ranges. In addition, runs between Q1 and Q3 minimized the effect of wind conditions prevailing at the time of the experimentation. The ships normally work either with or against the sea so that running lines this way was closer to real operating conditions than running lines over the whole area.

In view of the relatively rapid rate at which environmental conditions can change it was decided to place a standard target (sonar reflector) at the origin and to require each run to pass over this target. In this way a maximum detection range was obtained for every run. This was used in the analysis as an indicator of prevailing sonar conditions. In order to obtain a reasonable spread of lateral ranges at which observations were made, the targets in each block were placed in an approximately square fashion. The objects in Q2 and Q4 were placed in a way which gave lateral ranges of respectively 70 to 205 metres and 130 to 305 metres. The objects in Q1 and Q3 gave lateral ranges of approximately 0 to 600 metres. The objects in each block were not placed in a precisely square fashion in order to make it more difficult for the sonar operator to detect a pattern in the layout. The target positions are shown in figure 3 above. The next step in planning the experimental design was to choose a set of tracks. Let θ denote the angle between the X-axis and a typical track as shown in figure 3.

Since we are only interested in tracks between Q1 and Q3, θ can vary from 0° to 90° . Each track generates 16 detection opportunities at 16 corresponding lateral ranges, some of which may be equal.

Using this information the tracks for the experiments were selected to give a reasonable spread of lateral ranges. The total number of tracks that could be run was limited by ship availability. The experiments were carried out using the tracks as shown in table 1.

In order to minimize any learning effect on the part of the sonar operators, the tracks were run in a random order. Note, however, that the

position of the sonar reflector was given to the sonar operator to see how early he could detect it. The sooner the reflector is detected, the better the sonar conditions.

Table 1: Angles of tracks run during experiments

SEQUENCE	ANGLE
1	80
2	80
3	50
4	50
5	35
6	55
7	40
8	25
9	5
10	30
11	5
12	45
13	40
14	60
15	50
16	45
17	40
18	45
19	45
20	25
21	45
22	50
23	30
24	45
25	40

Data obtained from this trial suggested that the lateral range functions for both object types can be approximated by a logistic regression model. Using the results of this trial (refer to table 2) as prior information and the criterion of *D*-optimality we now see if we can improve on the above design.

Remarks:

- (i) Several other factors had to be taken into account in the experimental planning. These are, amongst others, criteria for

selecting and alternating sonar operators, sonar settings, search speed, etc. In making decisions regarding these factors, the guiding principle was to specify values that were either normally used or values that would be used in an actual operation. Thus the decision was to use only fully trained operators and to change operators after every run (about 20 to 30 minutes). For the same reason it was decided to use a fixed sonar setting and search speed.

- (ii) As mentioned above, sonar conditions were monitored by requiring each run to pass over the standard target and measuring the maximum detection range. This was found to be a good indicator of sonar conditions and was used as a covariate in the analysis. Although we will not report on it, environmental conditions were further monitored by measuring water temperatures, salinity, winds and currents. These factors were used to explain variations in the maximum detection range.
- (iii) The results of this study were documented by de Wet et al. (1987).

4. DESIGN USING THE CRITERION OF D-OPTIMALITY

4.1 Introduction

Abdelbasit and Plackett (1983) considered the problem of optimal allocation for the values of the explanatory variable when the response is binary and assuming the logistic regression model. They used the criterion of local D-optimality and, assuming symmetry, showed that for even sample size the optimal design is a two-point design where the same number of observations is taken at each value of the explanatory variable. Because one needs to know the parameters to design an experiment optimally, they suggested a sequential approach. Minkin (1987) derived the optimal allocation for the local criterion without unnecessary restrictions and improved on the two-stage approach of Abdelbasit and Plackett. We also proceeded along these lines and used the criterion of local D-optimality to obtain optimal object positions. Subject to practical constraints, mentioned in section 3,

these optimal positions were adjusted to obtain new designs and the efficiency of these were compared to the optimum one. We now discuss the model that we used and the design strategy in more detail.

4.2 The model

Suppose we want to place n objects of a specific type in the experimental region and that we choose m different tracks running through the centre of the region, the j -th track making an angle θ_j with the X-axis. (See figure 4.)

Let (x_i, y_i) denote the position of the i -th object. Then the lateral range of the i -th object to the j -th track is given by

$$(3) \quad r_{ij} = r(x_i, y_i, \theta_j) = |y_i \cos \theta_j - x_i \sin \theta_j|$$

for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

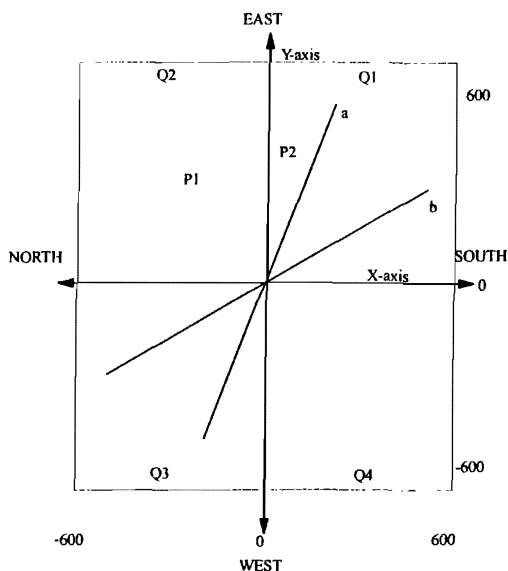


Figure 4: Experimental region indicating two tracks (a and b) and two objects (P_1 and P_2) and four quadrants (Q_1 , Q_2 , Q_3 and Q_4). Note that the positive X-axis points south.

Suppose we run along track j . Then we will have one detection opportunity for each object at its specific lateral range. Note that associated with each detection opportunity is a binary response variable having the value 1 when an object is detected and 0 otherwise. Suppose we replicate the experiment k times and let Z_{ijl} denote whether the i -th object was detected or not on the j -th track at replicate l . Then

$$(4) \quad R_{ij} = \sum_{l=1}^k Z_{ijl}$$

is the number of detections of the i -th object while on track j . Now R_{ij} has a binomial distribution with parameters k and

$$(5) \quad p_{ij} = P(Z_{ijl} = 1 | r_{ij}).$$

Assuming a logistic model we take

$$(6) \quad \text{logit } p_{ij} = \alpha + \beta r_{ij} \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, m$$

where α and β are parameters that have to be estimated from the data. We are only interested in values of $\alpha > 0$ and $\beta < 0$. The local D-optimality criterion is equivalent to maximizing the determinant of the Fisher Information matrix. Since the likelihood function is given by

$$(7) \quad \prod_{i=1}^n \prod_{j=1}^m \binom{k}{R_{ij}} p_{ij}^{R_{ij}} (1 - p_{ij})^{k-R_{ij}}$$

the determinant of the Fisher Information matrix follows easily as

$$(8) \quad k^2 \left\{ \sum_{i=1}^n \sum_{j=1}^m w_{ij} \sum_{i=1}^n \sum_{j=1}^m r_{ij}^2 w_{ij} - \left[\sum_{i=1}^n \sum_{j=1}^m r_{ij} w_{ij} \right]^2 \right\}$$

where $w_{ij} = p_{ij}(1 - p_{ij})$.

Minkin (1987) showed that when the total number of observations ($N = nmk$) is even, the above determinant is maximized (in our case) when

we choose the objects and tracks in such a way that half the observations are taken at a lateral range of $(-\alpha - 1,5434)/\beta$ and the other half at a lateral range of $(-\alpha + 1,5434)/\beta$ for $\alpha > 1,5434$. For $\alpha \leq 1,5434$ half the observations are taken at a lateral range of 0 and the other half at a range which is obtained by solving for r in

$$(9) \quad \beta r(1 - \exp(\alpha + \beta r)) / (1 + \exp(\alpha + \beta r)) = 2.$$

We now give you an example to illustrate some concepts needed in further discussion.

Example

The first experiment indicated that for metal objects α is in the region of 2,0 and β in the region of -0,025 (refer to table 2). We use these values for α and β in what follows. If only one track is run ($m=1$), local D-optimality with an even number of objects can be obtained from the above results. In this case, half the objects are placed anywhere on either one of two lines parallel to the track, at a distance of 18,26 metres and the other half anywhere on two lines at a distance of 141,74 metres. Note that for practical considerations previously mentioned, objects must be placed at least 100 metres from each other. In the case that two tracks are run ($m=2$), local D-optimality can still be obtained as follows. For each track, draw two parallel lines on either side, two at a distance of 18,26 metres and two at a distance of 141,74 metres. The four 18,26 metre lines lead to four intersections with one another and similarly for the four 141,74 metre lines. These eight points of intersection are given in figure 5.

Remarks

- (i) Although we have two groups of four intersections each, our practical constraint of 100 metres between objects, means that a maximum of four objects can be placed to obtain local D-optimality. Note that doing one replicate using this design will give the same information as four replicates of the one track design described above, containing two objects.

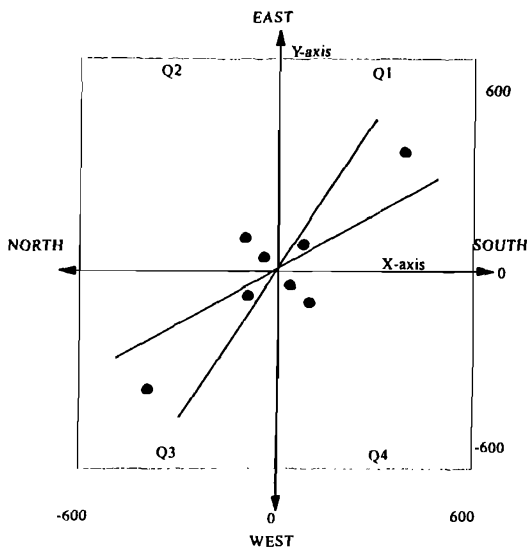


Figure 5: Local D-optimal design for two tracks

- (ii) A shortcoming of the above designs is that too few different tracks are run over the experimental region. It is known from experience that replicating a one or two track design would result in learning of the experimental layout by the operator. However, when we increase the number of tracks we can no longer place the objects in "D-optimal" positions. We can, however, get close to this by using the procedure described in the next subsection.

4.3 m-Track designs

Suppose we run $m(>2)$ tracks over the experimental region, the j -th track making an angle

$$(10) \quad \theta_j = 90 - j(90/(m+1)) \quad j = 1, 2, \dots, m$$

with the X-axis. From the discussion in the previous section we know that local D-optimally is obtained by placing an even number of objects in pairs at two "optimal" lateral ranges, say l_1 and l_2 . A reasonable approach would be to place the objects in positions having lateral ranges as close as possible to the optimal ones. Using as measure of closeness the minimum sum of squared differences we obtain a maximum of four positions as follows. Let (x_i, y_i) denote the i -th position, obtained by solving

$$(11) \quad \min_{x_i, y_i} \sum_{j=1}^m (r(x_i, y_i, \theta_j) - l_i)^2 \quad \text{for } i = 1, 2$$

with $r(x_i, y_i, \theta_j)$ given in (3). Using (3) and (10) we obtain

$$(12) \quad \begin{aligned} y_i &= \pm l_i \left(\sum_{j=1}^m \sin \theta_j \right) / \left(\sum_{j=1}^m \sin \theta_j \cos \theta_j + \sum_{j=1}^m (\sin \theta_j)^2 \right) \\ x_i &= \mp l_i \left(\sum_{j=1}^m \sin \theta_j \right) / \left(\sum_{j=1}^m \sin \theta_j \cos \theta_j + \sum_{j=1}^m (\sin \theta_j)^2 \right) \end{aligned}$$

for $i = 1, 2$.

Note that this procedure yields two positions in Q_2 and two in Q_4 all lying on a line making an angle of 45° with the negative X-axis. In the example below we will show that this procedure yields designs having high efficiency when compared to the local D-optimal design. As measure of efficiency we use the ratio of the determinant of the Fisher Information matrix for the design of interest to the determinant of the Fisher Information matrix for a local D-optimal design, times 100. Here the local D-optimal design used is as described in the previous section having the same number of detection opportunities.

For the range of α - and β -values considered it can often happen that the distance between the closest objects is less than 100 metres.

Example

As in the previous example we consider metal objects. If three tracks are run the above procedure yields four positions. (Two in Q_2 and their reflections in Q_4). The two positions in Q_2 are $(-13,59; 13,59)$ and $(-105,43; 105,43)$ giving an efficiency of 99,77% with respect to the local D-optimal design. For six tracks we obtain the position of the two objects as $(-13,86; 13,86)$ and $(-107,52; 107,52)$ giving an efficiency of 99,2%. (Remember these efficiencies are calculated at $\alpha = 2,0$ and $\beta = -0,025$).

Remarks

- (i) From a practical viewpoint, three tracks were considered the lower limit in order to guard against the possibility of the operator learning effect mentioned earlier. On the other hand, bearing in mind that the tracks should run along a north-west south-east direction, six tracks were considered a reasonable upper limit from a navigational stance.
- (ii) For the non-metal objects (where typically α is in the region of 0,2 and β is in the region of -0,03) we found the above procedure yielded positions too close to that for the metal objects, meaning that only two objects of a specific type can be placed.
- (iii) We have calculated efficiencies for values of α and β in the range of interest to us and throughout found them to be greater than 99%. An interesting point is that the efficiencies stay constant over α and only depend on β .

Although the above procedure results in designs having an extremely high efficiency with respect to the local D-optimal one, it suffers from a serious drawback in that too few objects can be placed. Ideally we want to place at least two objects of each type in the region and run at least three different tracks in order to reduce the learning effect and to estimate simultaneously the lateral range curves for the two object types. To this end we did a grid search which is described next.

4.4 Grid search

Assuming a minimum distance of 100m between objects we marked possible object positions as indicated on the grid in figure 6. For the three- and six-track designs described in the previous section we proceeded as follows to obtain the object positions giving highest efficiency.

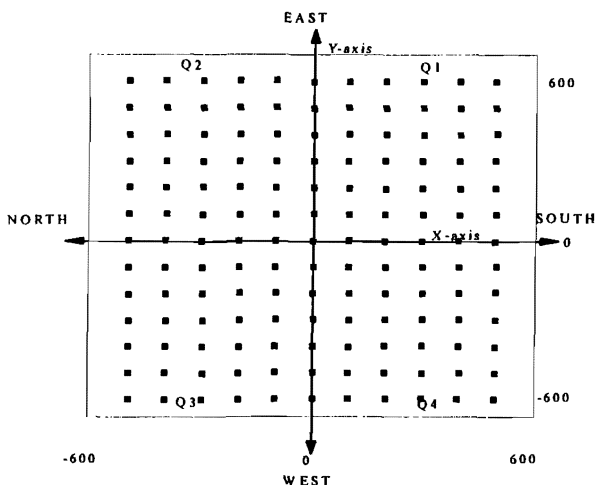


Figure 6: Grid of experimental region

For every combination of two positions we calculated the efficiency of the design with respect to the local D-optimal one. (We searched in combinations of twos because it seemed natural from the discussion in the previous two subsections.) We did this for each object type and over the ranges of α - and β -values considered. Table 2 contains the values of α and β . Here the lower and upper values roughly correspond to 99% confidence limits obtained in the first experiment.

Table 2: Range of α - and β -values

	lower	expected	upper	lower	expected	upper
METAL	1,8	2,0	2,2	-0,02	-0,025	-0,03
NON-METAL	0,0	0,2	0,4	-0,025	-0,03	-0,035

We then calculated the minimum efficiency of each combination over the range of α - and β -values considered and then chose the object positions from those combinations having a minimum efficiency of at least 70%. Using this procedure we arrived at the following design for three tracks. Four metal objects are placed at (200; 100), (300; 300), (-200; -100) and (-300; -300) having a minimum efficiency of 73,1%. Four non-metal objects are placed at (0; 0), (0; 100), (-100; -100) and (200; 200) having a minimum efficiency between 78,2% and 76,3%.

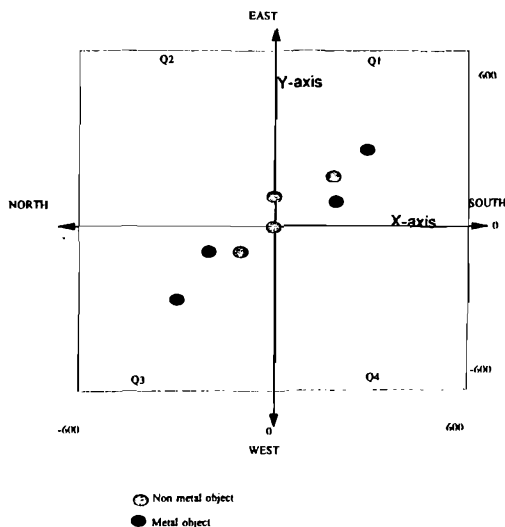


Figure 7: Object layout for three-track design for metal objects and non-metal objects

Similarly for six tracks the design is depicted in figure 8. Here we place two metal objects at $(-100; -200)$ and $(200; 100)$ having minimum efficiency of 72,3%. Two non-metal objects are placed at $(0; 0)$ and $(0; -100)$ having minimum efficiency of 78%.

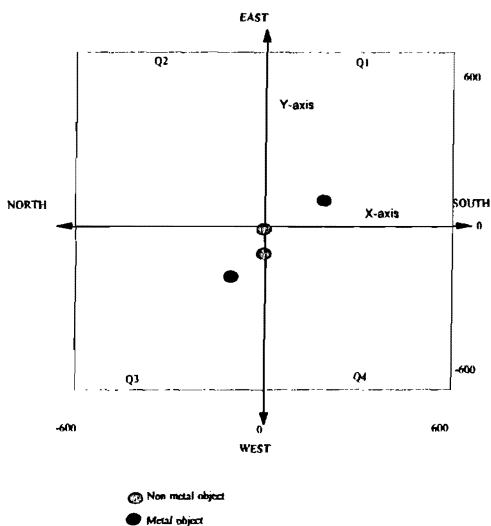


Figure 8: Object layout for six-track design for metal objects (M) and non-metal objects (N)

Note that both these designs will result in 24 detection opportunities for one replicate. When comparing efficiencies of the joint three- and six-track designs, no clear choice seems evident. We settled for the object layout in the three-track design, replacing the non-metal object at $(0; 100)$ with $(-75; 75)$. The latter position resulted in higher efficiency and was derived using the procedure in section 4.3. In order to measure sonar conditions two sonar reflectors are placed at $(0; -100)$ and $(100; 0)$ which are close to the initially

planned position, viz $(0; 0)$. The final layout of the objects is depicted in figure 9. Note that this design is preferable to the D-optimal ones as far as model validation is concerned.

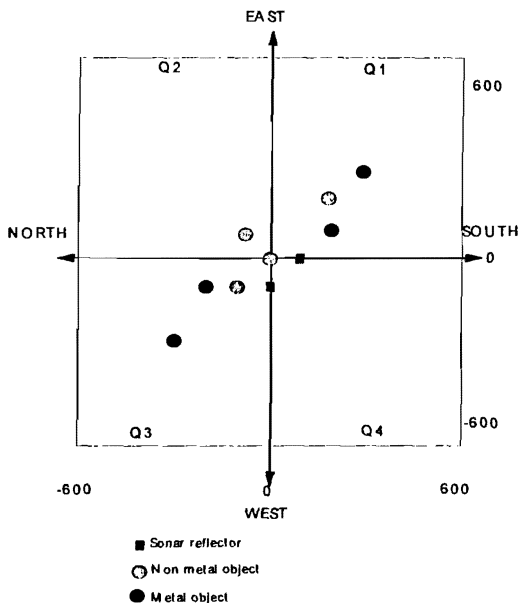


Figure 9: Final design

Using the results of the first experiment as our basis we calculated the efficiencies of the final proposed experimental layout (figure 9) and that of the initial experimental layout (figure 3) over the range of α - and β -values considered. The resulting efficiencies appear in table 3.

Table 3: Comparison of efficiencies between initial and final design

OBJECT TYPE	α -value	β -value	INITIAL DESIGN	FINAL DESIGN
Non-metal	0,0	-0,035	8,3%	84,6%
		-0,030	10,9%	86,8%
		-0,025	14,9%	84,3%
	0,2	-0,035	8,8%	84,9%
		-0,030	11,6%	86,6%
		-0,025	15,8%	83,4%
	0,4	-0,035	9,4%	85,3%
		-0,030	12,3%	86,4%
		-0,025	16,9%	82,3%
Metal	1,8	-0,030	22,5%	73,1%
		-0,025	34,1%	88,6%
		-0,020	49,0%	90,9%
	2,0	-0,030	24,9%	76,5%
		-0,025	37,5%	91,0%
		-0,020	52,9%	89,3%
	2,2	-0,030	27,3%	79,3%
		-0,025	40,9%	91,9%
		-0,020	56,4%	85,8%

Clearly, the final design is a vast improvement on the initial design in a D-optimal sense.

5. CONCLUDING REMARKS

We have proposed an experimental design for evaluating sonar detection performance. The final design consisted of three search tracks over the experimental area and specified the position of four objects of each type together with two sonar reflectors. The latter serve as reference points and as indicators of the prevailing sonar conditions.

This paper provides a case study of a real world experimental design (with distinctly unique requirements) which demonstrates the benefits of collaboration between experimenters and statistics, specifically experimental design, within the broader context of decision making. Perhaps the most important aspect of this design is that it satisfies the needs of the experimenters.

The proposed design is not a traditional text book design. It is not optimal in any statistical or other sense, sophisticated mathematics has not been used and it had to meet many practical constraints. However, design methodology and local D-optimality have been kept sharply in focus as useful reference points. Prior knowledge has been used, although not in a formal Bayes sense, and minimax (or maximin) ideas have been employed yet the design is not minimax. The continuous design space has been discretised for computer search. Although the design is a special case, experimental design methodology provided the foundation on which alternative designs could be meaningfully compared.

The work described in this paper has again emphasized to us the important point that in real world experimental design the reasoning process, together with the collaboration, is the most instructive part rather than the final design. Here we have endeavoured to guide the reader through this reasoning process and have especially emphasized practical constraints and considerations along the way. Our experience certainly corroborates the views about experimental design in the real world that have been so eloquently expressed by a host of authors.

ACKNOWLEDGEMENT

The initial experiment referred to in this paper was carried out jointly with Prof. F. Lombard and Dr. J.P. Strümpfer. Their valuable contribution to this work is gratefully acknowledged.

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